Trajectory Clustering and Dynamic In-Context Prompting for Adaptive Goal-Oriented Conversational AI

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Project Repository & Data Availability: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

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1.1 Background and Problem Domain

Modern conversational artificial intelligence systems are increasingly deployed across high-stakes, consultative customer service and travel domains such as online booking, hospitality advisement, and customer experience operations (Jannach et al., 2021). Despite significant breakthroughs in large language model (LLM) architectures, state-of-the-art commercial agents suffer from a fundamental behavioral limitation: they rely almost exclusively on static, one-size-fits-all system prompts. Consequently, the agent exhibits rigid, monolithic conversational pacing, identical tonal characteristics, and invariant elicitation strategies regardless of whether the interacting human is an exploratory, hesitant first-time buyer requiring consultative reassurance or a time-sensitive, transactional corporate traveler seeking instantaneous booking confirmation (Inaba et al., 2022). This structural misalignment between user personality archetype and system conversational pacing leads to user friction, repetitive clarifying turns, and premature session abandonment.

Simultaneously, when practitioners attempt to adapt conversational style dynamically through prompt engineering, they encounter severe factual hallucination and policy drift (Rebedea et al., 2023). Injecting diverse, unconstrained conversational exemplars into the LLM context window often causes the model to leak stale constraints, hallucinate pricing or dates from the exemplar dialogue, and violate non-negotiable operational business policies (e.g., booking without valid passport lead time or exceeding discount thresholds). There is an urgent empirical need to formulate a data-driven framework that bridges unsupervised dialogue clustering,

dynamic few-shot prompt conditioning, and strict declarative rule enforcement to achieve personalized, style-convergent, yet factually grounded conversational interactions.

1.2 Theoretical Foundation and Prior Art (MultiWOZ 2.2, Frames, S-BERT)

This study builds directly upon foundational literature in dialogue state tracking, sentence representation learning, and in-context learning. The primary theoretical baseline is established by the MultiWOZ benchmark (Budzianowski et al., 2018), specifically the MultiWOZ 2.2 release (Zang et al., 2020), which standardized dialogue act schemas, cleaned state annotation errors, and introduced schema-guided dialogue ontology definitions. To model multi-intent, negotiation-heavy travel interactions where customers dynamically revise constraints across turns, this study integrates the Microsoft Frames corpus (El Asri et al., 2017). Frames formalizes frame tracking, where conversational agents maintain multiple competing candidate itineraries as users explore trade-offs between flight dates, budget limits, and cabin classes.

In the representation domain, we leverage Sentence-BERT (Reimers & Gurevych, 2019) to construct dense semantic trajectory embeddings. Whereas traditional dialogue analysis relies on bag-of-words or turn-level surface lexical matching, S-BERT utilizes siamese and triplet networks to map entire conversational turn sequences into semantically meaningful vector spaces where geometric distance reflects structural conversational behavior. Finally, this research extends the in-context few-shot learning paradigm formalized by Brown et al. (2020) and refined by Liu et al. (2022). Liu et al. demonstrated that LLM downstream task performance is heavily governed by the semantic proximity of retrieved in-context exemplars. We advance this theory by demonstrating that in-context exemplars act as powerful stylistic conditioning priors that steer conversational tone, pacing, and turn economy without requiring parameter fine-tuning.

1.3 Problem Statement (Independent and Dependent Variables)

Existing goal-oriented conversational systems lack data-driven mechanisms to automatically discover latent user conversational archetypes from historical dialogue logs and adapt their interaction style in real-time, resulting in conversational friction, redundant elicitation, and high abandonment. Furthermore, unconstrained prompt-based style steering introduces factual leakage and policy drift that commercial platforms cannot tolerate.

In a quantitative data analytics framework, this research evaluates how key Independent Variables—specifically, Clustering Algorithm, In-Context Conditioning Strategy, Business Rule Enforcement Mode, and In-Context Exemplar Count—statistically impact primary measurable Dependent Variables: Structural Cluster Quality, Semantic Style Convergence, Factual Slot Accuracy, Exemplar Factual Leakage Rate, Operational Rule Adherence Rate, and Conversational Efficiency. Detailed operational definitions, metric scales, and analytical roles for each variable are formalized in Table 2.

1.4 Purpose of the Study

The purpose of this quantitative data analytics study is fourfold: (a) to discover, evaluate, and interpret latent conversational archetypes from human-human travel dialogues using dense trajectory embeddings; (b) to quantify the degree of semantic and stylistic convergence achieved by dynamic cluster-adaptive few-shot prompt injection compared to static baselines; (c) to conduct a controlled 2x2 ablation experiment isolating the causal impact of declarative business rules versus few-shot conditioning on factual slot preservation and policy adherence; and (d) to evaluate conversational efficiency gains across distinct traveler archetypes. The investigation delivers a rigorous, fully reproducible methodology for context-adaptive conversational AI.

1.5 Scope and Analytical Boundaries

The scope of this research is strictly bounded to open-access, multi-turn English-language goal-oriented dialogue benchmark datasets, specifically the Microsoft Frames corpus (1,369 dialogues) and the MultiWOZ 2.2 benchmark (10,438 dialogues). The investigation utilizes dense sentence representation models (Sentence-BERT all-MiniLM-L6-v2) to vectorize conversational turn trajectories and evaluates distance- and density-based clustering algorithms operating in Euclidean and cosine metric spaces. By relying exclusively on published, peer-reviewed public corpora, the methodology guarantees 100% scientific reproducibility without proprietary data dependencies. Operational implementation schedule details and syllabus milestone alignments are documented in Appendix B.

1.6 Research Questions (RQ1–RQ4)

To guide the investigation, four focused research questions are established:

• Research Question 1 (RQ1 - Unsupervised Trajectory Clustering Separation): Which unsupervised clustering algorithm (K-Means, Agglomerative Hierarchical, or HDBSCAN) operating on dense trajectory embeddings produces the highest structural cluster separation (Silhouette Score and Davies-Bouldin Index) and distinct behavioral archetypes across multi-turn goal-oriented dialogues?

• Research Question 2 (RQ2 - Dynamic In-Context Semantic Style Convergence): To what extent does dynamic cluster-adaptive few-shot exemplar injection increase the agent's Semantic Style Closeness (Cosine Similarity and BERTScore to target reference cluster centroids) compared to zero-shot and static few-shot baselines?

• Research Question 3 (RQ3 - Controlled Ablation: Rules vs. Style Factual Drift): In a controlled 2x2 ablation experiment (Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]), does aggressive stylistic convergence induce Exemplar

Factual Leakage or degrade Slot F1, and does a decoupled declarative rule engine eliminate policy violations without dampening style convergence?

• Research Question 4 (RQ4 - Turn Economy & Friction Across Archetypes): How does cluster-adaptive few-shot prompting impact conversational efficiency—measured via Turns to Goal Completion (TGC) and Redundant Elicitation Rate (RER)—when interacting across distinct customer conversational archetypes (e.g., Consultative/Ambiguous vs. Direct/Transactional)?

2. Research Methodology, Randomization, and Sample Size Determination

2.1 Method Selection, Golden Partitioning, and Power Rationale

To ensure robust statistical validity and prevent Type I and Type II errors, sample size calculations were performed for each research question. One-Way Analysis of Variance (ANOVA) power analysis was selected for RQ1 and RQ2 to test differences across clustering algorithms and prompt conditioning treatments. A Two-Way Factorial ANOVA power calculation was utilized for the 2x2 ablation design in RQ3. For RQ4, the Confidence Interval method for population means was applied to estimate dialogue efficiency metrics within a tight margin of error. Across all tests, standard statistical thresholds are enforced: significance level alpha = 0.05 (two-tailed, $Z_{alpha/2} = 1.96$) and statistical power 1 - beta = 0.80 ($Z_{beta} = 0.8416$).

Train-Test Separation and Golden Reference Architecture: In quantitative machine learning and conversational analytics, rigorous separation between training and evaluation partitions is critical to prevent data contamination (data leakage). To establish absolute experimental validity, the primary Microsoft Frames benchmark (1,369 dialogues) is formally partitioned into two mutually exclusive, disjoint subsets: (1) The Golden Reference and Retrieval

Pool (D_gold, 70%, N_train approx 958 dialogues): In this study, the training partition operationalizes the empirical 'Gold Standard'. Dialogues in this partition are rigorously filtered for full task success (saturated Tier-1 mandatory slots, verified booking commitment, zero mid-session abandonment). These Wizard-of-Oz conversations serve as the gold standard ground truth of high-performing human consultative dialogue, capturing optimal turn economy, cooperative negotiation, and domain compliance. Operationally, D_gold performs two foundational training functions: first, it is used exclusively to train unsupervised clustering algorithms (K-Means, Agglomerative, HDBSCAN) and establish canonical archetype reference centroids {C_1, C_2, C_3}; second, it populates the FAISS vector database from which k = 3 demonstration exemplars are dynamically retrieved. Crucially, the frozen centroids {C_1, C_2, C_3} derived from D_gold establish the fixed ground-truth reference coordinates against which Trajectory Deviation (delta = 1 - cos(v_generated, C_target)) and Semantic Style Convergence are subsequently measured. (2) The Held-Out Evaluation Test Cohort (D_test, 30%, N = 390 dialogues): In accordance with standard machine learning protocols, this evaluation cohort is held completely unseen during training, clustering, and centroid calculation. Drawn disjointly from Microsoft Frames, this test cohort matches the minimax sample size requirement derived in Section 2.3. All evaluation experiments for RQ1–RQ4 are executed strictly on D_test (where D_test intersect D_gold = empty set). When the LLM agent interacts with an unseen test scenario, the dynamic prompt assembler queries FAISS to retrieve k = 3 stylistic exemplars from D_gold without exposing test dialogue labels or trajectories. This architectural separation guarantees zero data leakage: the agent's generalization, factual slot preservation, and policy adherence are evaluated strictly on novel test data while its conversational style is anchored against frozen Gold Standard exemplars.

Randomization and Bias Mitigation Strategy: To prevent selection, temporal, and ordering biases, dialogue instances across all corpora are subjected to uniform random shuffling prior to experimental partitioning. Trajectory evaluation splits are generated using Stratified ShuffleSplit cross-validation across 10 repeated iterations with fixed pseudorandom seeds. This randomization guarantees equal archetype representation, eliminates ordering and recency artifacts, and ensures findings reflect true behavioral patterns rather than dataset sequencing.

2.2 Stepwise Sample Size Calculations (RQ1–RQ4)

Stepwise Calculation for RQ1 (One-Way ANOVA - Cluster Quality Differences)

RQ1 compares Silhouette Scores across three clustering algorithms (K-Means, Agglomerative, HDBSCAN). To detect a medium standardized effect size of Cohen's f = 0.25 at alpha = 0.05 and power = 0.80 across k = 3 groups, Cohen's formula is applied: degrees of freedom df_groups = k - 1 = 2; critical F_(0.05, 2, inf) = 3.00; total sample size is given by N_total = lambda / f^2, where non-centrality parameter lambda = 9.64 for alpha = 0.05, power = 0.80, k = 3. Step 1: Compute f^2: (0.25)^2 = 0.0625. Step 2: Divide lambda by f^2: 9.64 / 0.0625 = 154.24. Step 3: Divide by 3 groups: 154.24 / 3 = 51.41 per algorithm. Rounding up yields n = 52 dialogue samples per group (N_RQ1 = 156 total cluster evaluation units).

Stepwise Calculation for RQ2 (One-Way ANOVA - Semantic Style Convergence)

RQ2 assesses Semantic Cosine Similarity across three conditioning strategies (Zero-Shot, Static Few-Shot, Cluster-Adaptive Few-Shot). To detect a moderate effect size of Cohen's f = 0.28 (reflecting anticipated prompt steering differentials) with alpha = 0.05 and power = 0.80 across three groups: Step 1: Compute f^2: (0.28)^2 = 0.0784. Step 2: Divide lambda = 9.64 by 0.0784: 9.64 / 0.0784 = 122.96. Step 3: Divide by 3 groups: 122.96 / 3 = 40.98 per group.

Rounding up yields n = 42 dialogues per treatment arm (N_RQ2 = 126 total dialogue evaluations).

Stepwise Calculation for RQ3 (Two-Way Factorial ANOVA - 2x2 Ablation Experiment)

RQ3 evaluates the 2x2 ablation design (Rules [Active vs. OFF] x Prompting [Zero-Shot vs. Cluster Few-Shot]) on Factual Slot F1 and Rule Adherence Rate. For a 2x2 factorial structure (4 cells), to detect a main effect or interaction effect of size f = 0.25 at alpha = 0.05 and power = 0.80: degrees of freedom df_num = 1; critical F_(0.05, 1, inf) = 3.84; required non-centrality lambda = 7.85. Step 1: Total sample size N_total = lambda / f^2 = 7.85 / 0.0625 = 125.6 dialogues. Step 2: To account for potential dialogue truncation or incomplete synthetic runs, an operational buffer of 40% is incorporated: 125.6 * 1.40 = 175.84. Step 3: Allocating equally across 4 factorial cells yields n = 45 dialogues per cell. Total required sample size is N_RQ3 = 4 * 45 = 180 multi-turn dialogues.

Stepwise Calculation for RQ4 (Confidence Interval Method - Turn Economy Estimation)

RQ4 estimates the mean Turns to Goal Completion (TGC) and Redundant Elicitation Rate (RER) within a precise margin of error E. Based on prior dialogue corpus literature (El Asri et al., 2017), dialogue length exhibits standard deviation sigma = 3.5 turns. To achieve a narrow 95% confidence interval margin of error E = 0.35 turns (representing 10% of a standard deviation) at Z_(alpha/2) = 1.96: Step 1: Compute numerator: (Z_(alpha/2) * sigma)^2 = (1.96 * 3.5)^2 = (6.86)^2 = 47.0596. Step 2: Compute denominator: E^2 = (0.35)^2 = 0.1225. Step 3: Divide numerator by denominator: 47.0596 / 0.1225 = 384.16. Rounding up yields N_RQ4 = 385 dialogue interactions.

2.3 Sample Size Summary and Minimax Decision (N = 390)

To ensure all four research questions achieve full statistical validity without under-powering any subgroup analysis, the final sample size decision follows the formal minimax decision rule: N_final = max(N_RQ1, N_RQ2, N_RQ3, N_RQ4) = max(156, 126, 180, 385) = 385 dialogues. To facilitate balanced cross-validation across 3 cluster archetypes and 4 experimental folds, the sample size is rounded to exactly N = 390 dialogues (providing 130 dialogues per behavioral archetype). This cohort constitutes the 30% held-out test split drawn from Microsoft Frames, guaranteeing full statistical power using public data alone. This final decision is highlighted in Table 1 below.

Table 1

Sample Size Determination Summary and Final Minimax Decision across Research Questions

RQ #	Analytical Focus	Method	Alpha	Power	Target f / d	Required N
RQ1	Cluster Separation (Silhouette)	One-Way ANOVA	0.05	0.80	f = 0.25	N1 = 156
RQ2	Style Convergence (Cosine Sim)	One-Way ANOVA	0.05	0.80	f = 0.28	N2 = 126
RQ3	2x2 Ablation (Rules vs Prompting)	2x2 Fact. ANOVA	0.05	0.80	f = 0.25	N3 = 180
RQ4	Turn Economy (Turns to Goal)	Confidence Int.	0.05	N/A	E = 0.35, s=3.5	N4 = 385
Final	**Comprehensive Study Cohort**	**Minimax Decision**	**0.05**	**>= 0.80**	**Max(RQ1-RQ4)**	**N = 390**

Note. Final sample size N = 390 satisfies the minimax criterion N = max(N_RQ1..N_RQ4) and enables equal partition across 3 cluster archetypes.

3. Data Infrastructure, Two-Tier Ontology, and Project Repository

3.1 Primary Open Research Datasets and Access Links

To guarantee 100% academic reproducibility, this research utilizes widely recognized, open-access multi-turn conversational corpora. The primary training and clustering dataset is the

Microsoft Frames corpus (El Asri et al., 2017), comprising 1,369 human-human dialogues (19,986 individual turns) collected through a Wizard-of-Oz interface. In Frames, users search for vacation packages while exploring complex constraints (dates, destination trade-offs, budget adjustments, and cabin classes). The corpus is publicly accessible at: https://www.microsoft.com/en-us/research/project/frames-dataset/

For schema validation and dialogue state tracking cross-evaluation, we incorporate MultiWOZ 2.2 (Zang et al., 2020), which provides 10,438 multi-domain dialogues spanning hotel, attraction, flight, and taxi domains. MultiWOZ 2.2 features standardized slot definitions and corrected dialogue state labels, accessible at: https://github.com/budzianowski/multiwoz

3.2 Two-Tiered Canonical Slot Schema (Meta-Ontology Alignment)

A central methodological challenge when combining diverse conversational datasets is ontology divergence. Microsoft Frames uses slot labels such as or_city, dst_city, str_date, and budget, whereas MultiWOZ employs hotel-pricerange, flight-arriveBy, and taxi-departure. Furthermore, operational travel agents require policy constraints (e.g., passport validity, advance booking windows) absent from raw conversational text. To resolve this, we adopt the meta-ontology alignment methodology formalized in MultiWOZ 2.2 (Zang et al., 2020) and construct a Two-Tier Canonical Slot Schema:

• Tier 1 (Universal Core Slots): Standardized across all datasets and models, including: Origin_City, Destination_City, Departure_Date, Return_Date, Duration_Nights, Traveler_Count (Adults/Children), Budget_Amount, Accommodation_Tier, and Booking_Status.

• Tier 2 (Operational Extension Slots): Policy-level operational parameters evaluated in RQ3/RQ4, including: Visa_Processing_Window_Days, Passport_Expiry_Months, Package_Type (Group/FIT), Sharing_Basis (Twin/Single), and Discount_Applied_Percentage.

This two-tier schema ensures seamless slot mapping across heterogeneous datasets while isolating operational policy rules from prompt formatting.

3.3 Cross-Corpus Reference Clustering: Frames and MultiWOZ Transfer

To validate whether conversational trajectory archetypes discovered in Microsoft Frames generalize across independent data distributions, we conduct cross-corpus domain adaptation against MultiWOZ 2.2 (travel and accommodation domains). Because Frames focuses on multi-turn constraint trade-offs whereas MultiWOZ models slot-directed task completion, this cross-corpus evaluation tests the structural stability of the learned latent space. To prevent spurious lexical clustering driven by entity tokens (specific hotel or city names), all named entities and currency expressions are normalized to generic tokens ([DESTINATION], [ORIGIN], [HOTEL_NAME], [CURRENCY]) prior to vectorization. Dialogues from MultiWOZ 2.2 are projected directly into the latent trajectory space established by Frames, evaluating whether the discovered behavioral archetypes (Hesitant Explorer, Budget Negotiator, Rapid Transactional) demonstrate geometric clustering invariance across distinct open-access corpora.

3.4 Ethical Considerations, Bias Mitigation, and Random Shuffling

Because conversational AI agents can inadvertently amplify demographic stereotypes or exhibit exclusionary behavior, rigorous ethical safeguards are enforced. First, data privacy and intellectual property are fully protected: this research utilizes exclusively open-access, anonymized public corpora (Microsoft Frames and MultiWOZ 2.2) distributed under permissive research licenses, completely eliminating confidentiality, corporate non-disclosure, or personal data leakage risks. Second, algorithmic bias in few-shot conditioning is actively mitigated. Cluster archetypes are defined purely on conversational interaction mechanics (interaction length, turn latency, constraint revision frequency) rather than user demographic attributes,

ensuring equitable agent behavior. Finally, uniform random permutation of dialogue logs is enforced across training splits to ensure cluster definitions remain invariant to arrival order.

3.5 Project GitHub Repository and Verification

To satisfy project reproducibility and open-science standards, a dedicated, accessible public GitHub repository has been established at: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. The repository houses all data ingestion scripts, Pydantic schemas, clustering pipelines, FAISS retrieval modules, declarative rule configurations, and automated evaluation harnesses. The complete repository directory structure, component schema, and execution commands are documented in Appendix A.

4. Comprehensive Data Dictionary

Table 2 provides the complete, tabular data dictionary defining all experimental variables, operational measures, data types, units, and methodological roles required for evaluating RQ1–RQ4.

Table 2

Comprehensive Experimental Data Dictionary for Goal-Oriented Dialogue Clustering and Adaptive Prompting

Variable Name	Data Type	Role	Operational Definition & Meaning	Units / Range	Relevance
dialogue_id	String	Identifier	Unique dialogue session code from Frames/MultiWOZ	Discrete ID	Unit of analysis for trajectory tracking
clustering_algo	Categorical	Independent	Algorithm: K-Means, Agglomerative, HDBSCAN	3 Categories	Primary factor for RQ1 cluster evaluation
conditioning_mode	Categorical	Independent	Prompting mode: Zero-Shot, Static Few-Shot, Cluster Few-Shot	3 Categories	Treatment factor for RQ2 style convergence

rule_engine_mode	Categorical	Independent	Ablation condition: 0=Rules OFF, 1=Declarative Rules Active	Binary {0, 1}	Treatment factor for RQ3 ablation experiment
target_archetype	Categorical	Independent	Assigned dialogue cluster (Explorer, Negotiator, Transactional)	3 Clusters	Stratification factor across RQ2–RQ4
exemplar_count_k	Integer	Independent	Number of in-context demonstration dialogues injected	Count (k in {1, 3, 5})	Evaluates demonstration density effects
silhouette_score	Float	Dependent	Measure of cluster cohesion and separation	Score [-1.0, 1.0]	Primary metric for RQ1 evaluation
davies_bouldin_idx	Float	Dependent	Ratio of within-cluster distance to separation	Index [0.0, inf)	Validation metric for RQ1 evaluation
cosine_similarity	Float	Dependent	Cosine similarity between generated turn and cluster centroid	Score [-1.0, 1.0]	Primary style convergence metric for RQ2
bert_score_f1	Float	Dependent	Token-level semantic similarity to target reference exemplars	Score [0.0, 1.0]	Secondary semantic alignment metric for RQ2
slot_f1_score	Float	Dependent	Harmonic mean of slot extraction precision and recall	Ratio [0.0, 1.0]	Factual slot preservation metric for RQ3
factual_leakage_rate	Float	Dependent	Proportion of hallucinated entities traced to exemplar prompt	Ratio [0.0, 1.0]	Direct measure of prompt contamination in RQ3
rule_adherence_rate	Float	Dependent	Proportion of turns respecting visa, pax, and blackout rules	Ratio [0.0, 1.0]	Primary compliance metric for RQ3 ablation
turns_to_goal	Integer	Dependent	Total turns required to resolve user goal successfully	Turns (1–25)	Primary conversational efficiency metric for RQ4
redundant_elicit_rate	Float	Dependent	Proportion of turns re-asking established constraints	Ratio [0.0, 1.0]	Dialogue friction metric for RQ4
budget_gap_closure	Float	Dependent	Proportion of initial budget discrepancy resolved	Ratio [0.0, 1.0]	Commercial negotiation metric for RQ4

Note. Variables span trajectory embeddings, prompt conditioning modes, ablation indicators, and evaluation metrics.

5. Analytic Approach and Solution to RQ1–RQ4

This section maps each of the four research questions to its rigorous analytic methodology, detailed technical feature mechanics, mathematical formulations, underlying assumptions, and expected empirical outputs. To ensure complete alignment with committee requirements, all open methodological decisions are formally documented as explicit empirical assumptions.

5.1 Solution to RQ1: Unsupervised Trajectory Clustering Quality

5.1.1 Trajectory Construction Mechanics and Mathematical Formulation

A foundational contribution of this research is the transformation of multi-turn conversational interactions into dense structural trajectory vectors. Let a completed dialogue session D of length T turns be represented as an ordered sequence of user and assistant turn pairs: D = {(u_1, a_1), (u_2, a_2), ..., (u_T, a_T)}. Converting D into a fixed-length representation suitable for distance-based clustering requires capturing both semantic evolution and interaction structure without losing temporal progression.

First, each individual turn pair t is mapped into a d = 384-dimensional dense semantic vector using Sentence-BERT (all-MiniLM-L6-v2): e_t = S-BERT(u_t || ' [SEP] ' || a_t) in R^384. Second, because early turns typically define constraint discovery while terminal turns capture negotiation closure or resolution, an exponential temporal decay weighting function w_t is applied across turns: w_t = exp(lambda * t / T) / sum_{k=1}^T exp(lambda * k / T), where lambda = 0.75 is the temporal progression parameter governing trajectory sensitivity. Third, to capture non-lexical conversational mechanics (pacing, hesitation, negotiation friction), an auxiliary structural feature vector s_D in R^6 is computed: s_D = [ln(T), L_user_avg,

L_asst_avg, Delta_budget_mention, Turn_to_First_Offer, Constraint_Revision_Count], where L represents character length and Delta indicates pricing adjustment count. The final unified trajectory vector v_D is constructed via concatenation and L2-normalization: v_D = L2_Normalize([sum_{t=1}^T w_t * e_t] || StandardScaler(s_D)) in R^390. This trajectory vector captures the joint semantic-structural arc of the dialogue, enabling clustering algorithms to separate interactions based on customer behavioral pacing.

5.1.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses: Null Hypothesis H0 states there is no significant difference in structural cluster separation (Silhouette Score) across K-Means, Agglomerative Hierarchical, and HDBSCAN algorithms (mu_K = mu_Agg = mu_HDB). Alternative Hypothesis H1 states significant differences exist in structural cluster separation across algorithms, with density-based or agglomerative methods outperforming standard spherical K-Means (mu_HDB != mu_K). Analytic Method: A One-Way Analysis of Variance (ANOVA) followed by Tukey's Honestly Significant Difference (HSD) post-hoc test evaluates Silhouette Scores across 10 repeated 5-fold cross-validation bootstrap iterations (N = 156 evaluation units). Formulae: The Silhouette Score for trajectory i is: s(i) = (b(i) - a(i)) / max(a(i), b(i)), where a(i) is mean intra-cluster distance and b(i) is mean nearest-cluster distance. The Davies-Bouldin Index is: DB = (1/k) * sum_{i=1}^k max_{j != i} ((sigma_i + sigma_j) / d(c_i, c_j)). Methodological Assumptions (Addressing Committee Inquiries): (1) Normality & Homogeneity: Bootstrap Silhouette distributions satisfy normality and homoscedasticity (Levene's test p > .05). (2) Cross-Corpus Geometric Invariance: It is assumed that latent conversational trajectory structures discovered in Microsoft Frames represent universal human goal-oriented negotiation arcs (Hesitant Exploration, Budget Friction, Rapid Transactional). Consequently, dialogues from MultiWOZ 2.2 can be projected directly

into this latent embedding space via the identical S-BERT vectorizer without retraining embeddings, establishing cross-corpus validity across independent research benchmarks. Expected Outputs: Identification of optimal k = 3 behavioral archetypes (Hesitant Explorer, Budget Negotiator, Rapid Transactional) with Silhouette Score >= 0.42 and Davies-Bouldin Index <= 1.15.

5.2 Solution to RQ2: Dynamic In-Context Few-Shot Style Convergence

5.2.1 Exemplar Selection Logic, Centroid Triggers, and Vector Retrieval

To operationalize adaptive prompt steering during live multi-turn interactions, a precise exemplar retrieval trigger and matching pipeline is established. Exemplar selection operates via a four-stage real-time trigger mechanism: (1) Dynamic Retrieval Trigger: The exemplar selection mechanism is triggered at session initialization (turn t = 1) using the user's opening intent, and is re-evaluated at turn t = 2 upon receipt of the first constraint preference expression. The cumulative ongoing dialogue prefix P_t = {(u_1, a_1), ..., u_t} is vectorized in real time: q_t = S-BERT(sum_{i=1}^t w_i * Turn_i). (2) Centroid Proximity Matching: Query vector q_t is compared against precomputed cluster centroids {C_1, C_2, ..., C_K} established on the Golden Training Pool D_gold using cosine similarity: k* = argmax_{j in {1..K}} ((q_t . C_j) / (||q_t|| ||C_j||)). If max cosine similarity exceeds confidence threshold tau_conf >= 0.65, the interaction is assigned to target archetype k*. (3) FAISS Sub-Index Nearest-Neighbor Query: Within the isolated FAISS vector sub-index I_{k*}, an exact inner-product search retrieves the top k = 3 most structurally similar historical golden dialogues from D_gold: E = Top-3_{d in I_{k*}} ((q_t . d) / (||q_t|| ||d||)). (4) Slot-Masked Prompt Formatting: The retrieved golden dialogues undergo entity de-lexicalization (replacing specific dates, cities, and prices with canonical slot tokens) before being formatted into the Gemini Flash API system prompt prefix. This ensures the

LLM absorbs the conversational style, tone, and pacing of archetype k* without contaminating current dialogue state with stale entity values.

5.2.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses: Null Hypothesis H0 states dynamic cluster-adaptive few-shot conditioning does not significantly increase Semantic Cosine Similarity to the target reference cluster compared to static few-shot prompting (mu_cluster = mu_static). Alternative Hypothesis H1 states dynamic cluster-adaptive few-shot conditioning produces significantly higher Semantic Cosine Similarity and BERTScore to target reference cluster centroids (mu_cluster > mu_static). Analytic Method: A One-Way Repeated Measures ANOVA compares Cosine Similarity across three prompting treatments (Zero-Shot Baseline, Static Few-Shot, Dynamic Cluster Few-Shot) evaluated across N = 126 dialogue sessions (42 per condition) drawn from the held-out test cohort D_test. Formulae: Cosine Similarity between generated system response trajectory g and target cluster centroid c is: Cos_Sim = (g . c) / (||g|| * ||c||). BERTScore computes token-level greedy matching cosine similarity using contextualized RoBERTa embeddings. Methodological Assumptions (Addressing Committee Inquiries): (1) Sphericity & Independence: Sphericity is evaluated via Mauchly's test; if violated, Greenhouse-Geisser epsilon adjustments are applied. (2) Stylistic Prior Independence: It is assumed that in-context few-shot exemplars act purely as stylistic priors that guide turn pacing and language formality, while slot values and factual commitments remain under the control of the dialogue state tracker and business rule engine. Expected Outputs: Demonstration of statistically significant style convergence (Cosine Similarity increase from 0.58 baseline to >= 0.82 under dynamic conditioning, $p < .001$, Cohen's d >= 0.85).

5.3 Solution to RQ3: Controlled 2x2 Factorial Ablation of Rules vs. Style Factual Drift

5.3.1 Deterministic Goal Completion Criteria and Multi-Factor Success Rule

To rigorously quantify whether conversational style convergence introduces factual hallucination or compromises task success, a deterministic, unambiguous goal completion rule is enforced. A dialogue session D is classified as a binary success (Y_success = 1) if and only if all three Boolean criteria are simultaneously satisfied: (1) Mandatory Slot Saturation: All Tier-1 mandatory core slots (Origin_City, Destination_City, Departure_Date, Duration_Nights, Traveler_Count, Budget_Tier) must be fully populated in the final dialogue state tracker: State_slots intersect Required_slots = Required_slots; (2) Zero Unresolved Policy Violations: Automated programmatic evaluation against the declarative rule schema yields zero blocking violations (V_blocking = 0), verifying that no package was offered violating visa lead times, exceeding customer budget caps, or breaching passenger capacity limits; and (3) Explicit Terminal Commitment Act: The assistant successfully executes a formal commitment action (Offer_Package or Confirm_Itinerary) and the user responds with an explicit acceptance dialogue act (Inform(Accept) or Affirm()), concluding before max turns T <= 20. Mathematically: Success(D) = I((Slot_Saturation == True) and (V_blocking == 0) and (Terminal_Acceptance == True)). This deterministic definition eliminates subjective evaluation bias.

5.3.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses: Null Hypothesis H0 states active declarative rules do not significantly improve Rule Adherence Rate (RAR) or prevent factual leakage under cluster conditioning (Main Effect beta_rules = 0). Alternative Hypothesis H1 states active declarative rules significantly increase Rule Adherence Rate (RAR >= 0.98) and eliminate factual leakage, proving business rules causally enforce policy compliance without dampening style

convergence. Analytic Method: A Two-Way Factorial ANOVA (2x2 design: Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]) evaluates Slot F1, Exemplar Leakage Rate, and Rule Adherence across N = 180 dialogues (45 per cell) drawn from held-out cohort D_test. Formulae: Exemplar Factual Leakage Rate is: Leakage = (Hallucinated Entities Traced to Prompt Exemplars) / (Total Generated Entities); Rule Adherence Rate is: RAR = 1 - (Violations / System Turns). Methodological Assumptions (Addressing Committee Inquiries): (1) Decoupled Orthogonality Assumption: It is assumed that operational business rules can be completely decoupled into an external declarative configuration (rules_declarative.yaml) executed outside the LLM context window. The 2x2 factorial ANOVA assumes no unmodeled confounding between prompt structure and rule execution, allowing the main effect of rules and the interaction effect (Rules x Prompting) to isolate causal policy protection; and (2) Error Distribution: Factorial residuals meet assumptions of normality and homoscedasticity across the four balanced cells. Expected Outputs: Proof that declarative rules eliminate factual leakage (reducing leakage from ~14.2% in Rules-OFF down to 0.0% in Rules-Active) while maintaining high Slot F1 (>= 0.92).

5.4 Solution to RQ4: Dialogue Turn Economy and Friction Analysis

5.4.1 Declarative Rule Schema Scope and Precise Boundary Conditions

To establish reproducible boundaries for conversational efficiency and compliance, the declarative rule engine (rules_declarative.yaml) enforces four explicit classes of operational boundary conditions validated deterministically across all interactions: (1) Temporal and Visa Feasibility Boundaries: minimum booking lead time requires international departures >= 21 calendar days when visa processing is pending (domestic >= 48 hours), and international bookings require passport validity >= 6 months beyond the scheduled return date; (2) Financial

and Commercial Boundaries: maximum discount ceiling prohibits negotiated promotional discounts exceeding 10.0% of catalog base fare without manager escalation, and hard budget cap adherence strictly forbids proposing itineraries where Price > B_max; (3) Capacity and Operational Boundaries: minor accompaniment requires child travelers (< 18 years) to have >= 1 adult passenger (infant-to-adult ratio <= 1.0), and standard double room occupancy enforces a maximum of 3 adults per room (requiring an extra rollaway bed); and (4) Conversational Compliance Boundaries: opening turn must include organizational identity greeting, and cancellation policy and travel insurance disclaimers must be presented prior to terminal booking confirmation.

5.4.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses: Null Hypothesis H0 states cluster-adaptive prompting does not significantly reduce Turns to Goal Completion (TGC) or Redundant Elicitation Rate (RER) across archetypes (mu_cluster = mu_baseline). Alternative Hypothesis H1 states cluster-adaptive prompting significantly reduces TGC and RER, with the greatest turn reductions observed in complex consultative archetypes (mu_cluster < mu_baseline). Analytic Method: A Two-Way Factorial ANOVA evaluating Turns to Goal Completion across Conditioning Mode (3 levels) and Customer Archetype (3 levels) across N = 385 dialogue interactions drawn from held-out cohort D_test. Formulae: Redundant Elicitation Rate is defined as: RER = (Redundant Constraint Questions) / (Total System Turns), where a question is redundant if the requested slot is already saturated in state. Methodological Assumptions (Addressing Committee Inquiries): (1) Two-Tier Ontology Invariance: It is assumed that the Two-Tier Canonical Slot Schema harmonizes dataset schema differences between Microsoft Frames and MultiWOZ 2.2 without information loss, allowing turn economy metrics to be evaluated fairly across distinct public dialogue

distributions; and (2) Distributional Characteristics: Dialogue turn counts approximate a log-normal distribution; log-transformed turn counts are utilized in ANOVA models to guarantee normality. Expected Outputs: Significant reduction in dialogue turns (18–25% reduction in TGC for consultative users) and reduction in RER from 0.16 down to <= 0.03.

5.5 End-to-End System Integration and Evaluation Architecture

The experimental framework integrates all analytical components into an automated, reproducible machine learning evaluation pipeline. The pipeline executes four tightly coupled stages: (a) Trajectory Preprocessing and Vectorization, which parses Microsoft Frames and MultiWOZ raw dialogues into Tier-1/Tier-2 canonical slot trajectories and extracts 390-dimensional S-BERT embeddings with temporal decay weighting; (b) Unsupervised Archetype Indexing, which trains K-Means, Agglomerative, and HDBSCAN models to segment behavioral clusters and populates a localized FAISS vector index with D_gold exemplars; (c) Dynamic Prompt Assembler with Decoupled Verification, which matches incoming turns to centroids, retrieves top-k golden exemplars with slot masking, and interposes declarative assertions from rules_declarative.yaml; and (d) Automated Statistical Evaluation, which logs token-level Slot F1, BERTScore, Cosine Similarity, Exemplar Leakage Rate, and Turns to Goal Completion across randomized test splits. This integrated design ensures end-to-end experimental reproducibility without manual prompt tuning or proprietary database dependencies.

6. Practical Recommendations, Applications, and Stakeholder Value

6.1 Target Users and Beneficiaries

Beneficiaries span three key stakeholder groups: (1) Conversational AI Product Teams and AI Engineers, who gain an automated, data-driven method for persona steering without manual prompt re-writing; (2) Customer Experience and CRM Operations Leaders, who can

deploy bots that naturally mirror diverse customer pacing and tones, boosting engagement; and (3) NLP Researchers, who receive the first systematic empirical characterization of the trade-off between in-context stylistic convergence and factual hallucination leakage.

6.2 Actionable Recommendations and Business Impact

Organizations deploying LLM conversational agents should transition from static prompt engineering to Dynamic In-Context Clustering. Specifically: (a) cluster historical conversation logs to discover natural customer archetypes; (b) implement vector similarity nearest-neighbor lookup to supply style exemplars on-the-fly; and (c) enforce strict decoupled declarative rule engines to prevent exemplar factual leakage. In commercial customer service deployments, dynamic persona matching is projected to increase dialogue task completion by 20–30% while eliminating policy violations.

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Appendix A

Project Repository Architecture and Replication Guide

To ensure complete scientific reproducibility, all source code, schemas, and pipelines are maintained in a structured public GitHub repository. The project repository is accessible at:

https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

The repository follows standard open-source research layout as detailed below:

```
QM640-Dialogue-Clustering-Capstone/
├── data/
│   ├── raw/                  # Raw public corpora (Microsoft Frames, MultiWOZ 2.2)
│   ├── canonical/            # Canonicalized Tier-1 / Tier-2 mapped dialogue turns
│   ├── embeddings/           # Cached S-BERT 384-dimensional trajectory embeddings
│   └── multiwoz_eval/        # MultiWOZ 2.2 canonical cross-evaluation split
├── notebooks/
│   ├── 01_eda_and_cleaning.ipynb          # Exploratory data analysis & cleaning logs
│   ├── 02_trajectory_clustering_rq1.ipynb # RQ1 K-Means, Agglomerative, HDBSCAN
│   ├── 03_fewshot_retrieval_rq2.ipynb     # RQ2 FAISS vector indexing & Gemini inference
│   ├── 04_factorial_ablation_rq3.ipynb    # RQ3 2x2 Factorial ANOVA (Rules vs Drift)
│   └── 05_turn_economy_rq4.ipynb          # RQ4 Turn economy & friction statistics
├── src/
│   ├── ingestion.py          # MultiWOZ and Frames JSON parsers
│   ├── canonical_schema.py   # Pydantic schemas for Tier-1 and Tier-2 slots
│   ├── vectorizer.py         # Turn-level and trajectory Sentence-Transformers
│   ├── clustering.py         # Scikit-learn & HDBSCAN clustering pipelines
│   ├── faiss_index.py        # Exact L2/Cosine vector search for exemplar retrieval
│   ├── prompt_assembler.py   # Dynamic few-shot prompt construction with slot masking
│   ├── rule_engine.py        # Decoupled declarative business rule validator
│   └── evaluation.py         # Automated evaluators (BERTScore, Slot F1, Leakage)
├── rules/
│   └── rules_declarative.yaml # Declarative business constraints (visa, pax, budget)
├── outputs/
│   ├── figures/              # UMAP projections, Silhouette plots, interaction charts
│   └── tables/               # Exported ANOVA summary tables and CSV logs
├── requirements.txt          # Pinned Python package dependencies
├── .gitignore                # Ignored local docs, environment, and temp files
└── README.md                 # Project documentation and replication guide
```

Replication Instructions: Researchers can clone the environment and execute the pipeline via the following terminal commands: `git clone https://github.com/surnella/QM640-Dialogue-Clustering-Capstone.git && cd QM640-Dialogue-Clustering-Capstone && pip install -r requirements.txt && python src/ingestion.py && python src/vectorizer.py && python src/clustering.py`.

Appendix B

Ten-Week Phased Project Implementation Plan and QM640 Syllabus Milestone Alignment

To fulfill Walsh College QM640 requirements and align precisely with the official course schedule (Schedule- QM 640 - Sheet1), the project is organized across a realistic 10-week implementation lifecycle. All core computational data analytics, model engineering, and statistical hypothesis testing are completed by Week 8 ('Data Analysis Assessment'). Week 9 ('Data Analysis Refinement and Documentation') and Week 10 ('Document research results and finding') are dedicated to documentation refinement, presentation slide deck finalization, and formal capstone defense preparation. Week 11 provides post-submission refinement for minor committee adjustments with zero new analytical scope. Table B1 maps the official QM640 syllabus topics directly to our project milestones, analytical tasks, and deliverables.

Table B1

Ten-Week Phased Project Implementation Schedule, Deliverables, and QM640 Syllabus Milestone Alignment

Week & Dates	QM 640 Syllabus Topic	Capstone Phase & Analytical Tasks (Dialogue Clustering)	Tools & Stack	Course Deliverable & Status
Week 0 (Sep 26)	Introduction to Capstone	Problem statement selection and finalization; evaluate travel conversational AI vs clustering thesis	Git, Markdown, Python 3.11	Problem Statement Finalized
Week 1 (Sep 27 - Oct 3)	Define Research Problem and Topic	Scaffold GitHub repository; ingest raw Microsoft Frames & MultiWOZ 2.2 corpora; establish environment	Python, Pandas, GitHub, JSON	GitHub Repository Scaffolded
Week 2 (Oct 4 - Oct 10)	Synopsis Draft Submission	Formalize RQ1-RQ4; complete Two-Tier slot schema mapping; submit Synopsis Draft (Oct 6); live review (Oct 10)	Docx, Statsmodels, SciPy	Synopsis Draft Submitted (Oct 6)
Week 3 (Oct 11 - Oct 17)	Literature Search and Data Collection	Synopsis Final Submission (Oct 13); extract S-BERT 384d trajectory embeddings; generate EDA cleaning logs	S-BERT, Hugging Face, UMAP	Final Synopsis & Vectorized Corpus
Week 4	Formulate Objectives &	Interim Report Draft Submission (Oct 20); train baseline K-Means, Agglomerative,	Scikit-Learn, HDBSCAN,	Interim Report Draft

(Oct 18 - Oct 24)	Hypotheses	HDBSCAN (RQ1); live review (Oct 24)	Seaborn	(Oct 20)
Week 5 (Oct 25 - Oct 31)	Choose Research Design & Methodology	Interim Report Final Submission (Oct 27); construct FAISS sub-indexes for top-k exemplar retrieval; validate clusters	FAISS-CPU, Pydantic, YAML	Final Interim Report (Oct 27)
Week 6 (Nov 1 - Nov 7)	Data Analysis Plan	Final Report Draft Submission (Nov 3); implement Gemini Flash inference pipeline across Zero-Shot vs Dynamic Few-Shot (RQ2)	google-genai SDK, Gemini Flash	Final Report Draft & Generated Logs
Week 7 (Nov 8 - Nov 14)	Data Analysis Plan Execution	Execute 2x2 Factorial Ablation (Rules ON/OFF x Prompting); evaluate Slot F1, factual leakage, and RAR (RQ3)	Statsmodels, PyYAML, SQLite	RQ3 Ablation Matrix Completed
Week 8 (Nov 15 - Nov 21)	Data Analysis Assessment	**Complete Turn Economy (TGC) & Redundant Elicitation (RER) tests (RQ4); execute all One-Way/Two-Way ANOVAs; ALL ANALYTICS DONE**	Statsmodels, Pingouin, Matplotlib	Data Analysis 100% Completed
Week 9 (Nov 22 - Nov 28)	Data Analysis Refinement & Documentation	Capstone Final Presentation (Nov 27/28); Final Presentation Submission (Nov 29); documentation beautification & slide deck	PowerPoint, LaTeX, Seaborn	Final Presentation Slide Deck
Week 10 (Nov 29 - Dec 6)	Document Research Results and Finding	Author 15-page comprehensive capstone thesis; finalize GitHub documentation; Final Report Submission (Dec 13)	Word, GitHub, Zenodo	Final Capstone Thesis (Dec 13)
Week 11 (Dec 7 - Dec 13)	Refine Documentation of Results & Finding	Post-submission committee refinements; incorporate faculty feedback; final project archive (no new analytical scope)	GitHub Releases, PDF Archive	Final Archive Completed

Note. Implementation schedule perfectly matches the official QM640 schedule. All computational and statistical data analysis is 100% completed by Week 8 (Data Analysis Assessment), reserving Weeks 9–10 for documentation refinement, presentation delivery, and thesis defense.